

### Assignment 3 (Group\_41)

1. Determine the equilibrium point ( $x_{1e}$ ,  $x_{2e}$ ,  $u_e$ ,  $\alpha_e$ ) that corresponds to the condition when the vehicle is traveling with a velocity of 15 m/s on a flat road!

The calculated values are in MATLAB.

Problem 1:

At equilibrium all the derivatives are zero (steady-state)  
 $x_1 = x_{1e}$   $x_2 = x_{2e} = 15 \text{ m/s}$   $\alpha = \alpha_e = 0$  (flat road),  $u = u_e$

$x_{1e}$ : force at the wheel (equilibrium = 0)

$$\tau \frac{dx_1}{dt} = 0 \Rightarrow 0 = -x_{1e} + \frac{k r_{\text{gear}}}{r_w} \cdot u_e$$

$$\Rightarrow x_{1e} = \frac{k r_{\text{gear}}}{r_w} \cdot u_e \quad - (1)$$

$x_2$  = vehicle dynamics

$$m \frac{dx_2}{dt} = 0 \Rightarrow 0 = x_{1e} - mg \sin \alpha - \frac{1}{2} \rho C_D A_f x_{2e}^2 - mg \sin \alpha$$

$$\Rightarrow x_{1e} = mg \sin \alpha + \frac{1}{2} \rho C_D A_f x_{2e}^2$$

$$u_e = \frac{x_{1e} \times r_w}{k r_{\text{gear}}}$$

$$x_{2e} = 15 \text{ m/s}$$

$$\alpha_e = 0$$

2: Linearize the model around the equilibrium point  $(x_1e, x_2e, u_e, \alpha_e)$  the system matrices, A, B, C and H corresponding to the model:

$$\Delta \dot{x} = A \Delta x + B \Delta u + H \Delta \alpha$$

$$\Delta y = C \Delta x$$

problem 2.

$$\Delta \dot{x} = A \Delta x + B \Delta u + H \Delta \alpha$$

$$\Delta y = C \Delta x$$

To linearize the model around the equilibrium point.

$$\Delta x_1 = x_1 - x_{1e}$$

$$\Delta x_2 = x_2 - x_{2e}$$

$$\Delta u = u - u_e$$

$$\Delta \alpha = \alpha - \alpha_e$$

The force at the wheels.

$$\tau \frac{d\Delta x_1}{dt} = -\Delta x_1 + \frac{k r_{gear}}{r_w} \Delta u.$$

$$\frac{d\Delta x_1}{dt} = -\frac{1}{\tau} \Delta x_1 + \frac{1}{\tau} \frac{k r_{gear}}{r_w} \Delta u.$$

The vehicle velocity.

$$m \frac{d\Delta x_2}{dt} = \Delta x_1 - m g f \cos \alpha \Delta \alpha - \frac{1}{2} P C_D A_f x_{2e} \Delta x_2 - m g \sin \alpha \Delta \alpha.$$

$$\alpha = 0 \quad \cos(0) = 1 \quad \sin(0) = 0.$$

$\frac{1}{2} P C_D A_f x_{2e}^2$  is linearized around  $x_{2e}$

$$\frac{d\Delta x_2}{dt} = \frac{\Delta x_1}{m} - \frac{m g f \Delta \alpha}{m} - \frac{\frac{1}{2} P C_D A_f x_{2e} \Delta x_2}{m}.$$

$$\frac{d\Delta x_2}{dt} = \frac{\Delta x_1}{m} - g f \Delta \alpha - \frac{\frac{1}{2} P C_D A_f x_{2e} \Delta x_2}{m}.$$

$$\text{let } \Delta x = \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix}.$$

$$\frac{d\Delta x}{dt} = A \Delta x + B \Delta u + H \Delta \alpha.$$

$$= \begin{bmatrix} -\frac{1}{\tau} & 0 \\ \frac{1}{m} & -\frac{\frac{1}{2} P C_D A_f x_{2e}}{m} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix} + \begin{bmatrix} \frac{k r_{gear}}{\tau r_w} \\ 0 \end{bmatrix} \Delta u + \begin{bmatrix} 0 \\ -g f \end{bmatrix} \Delta \alpha.$$

$$\Delta y = C \Delta x. \quad (\text{output } y = x_2)$$

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

The matrix  $A = \begin{bmatrix} -\frac{1}{\tau} & 0 \\ \frac{1}{m} & -\frac{P_0 A_f x_{2e}}{m} \end{bmatrix}$  matrix  $B = \begin{bmatrix} \frac{k r_{gear}}{\tau r_w} \\ 0 \end{bmatrix}$

matrix  $H = \begin{bmatrix} 0 \\ -g t \end{bmatrix}$  matrix  $C = \begin{bmatrix} 0 & 1 \end{bmatrix}$

3. Determine the reachability matrix  $W_r$  of the linearized model! Is the system reachable?

problem 3.

The reachability Matrix  $W_r = [B \quad AB]$

The state space matrix  $A$  and  $B$  are

$$AB = \begin{bmatrix} -\frac{1}{\tau} & 0 \\ \frac{1}{m} & -\frac{P_0 A_f x_{2e}}{m} \end{bmatrix} \begin{bmatrix} \frac{k r_{gear}}{\tau r_w} \\ 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} -\frac{k r_{gear}}{\tau^2 r_w} \\ \frac{k r_{gear}}{m \cdot \tau r_w} \end{bmatrix}$$

$$W_r = [B \quad AB]$$

$$W_r = \begin{bmatrix} \frac{k r_{gear}}{\tau r_w} & -\frac{k r_{gear}}{\tau^2 r_w} \\ 0 & \frac{k r_{gear}}{m \cdot \tau r_w} \end{bmatrix}$$

The reachability matrix  $W_r$  has rank 2, which means that the system is reachable.

- 4: Design a state feedback cruise controller gain  $K$  using pole placement. The closed loop systems characteristic polynomial should be  $p(s) = s^2 + 2 \zeta \omega_n s + \omega_n^2$ , where  $\omega_n = 0.6$  and  $\zeta = 1/\sqrt{2}$

- The solution is provided in the MATLAB. Please refer Matlab.

$K = [-0.000018331318429, 0.342167704930532];$

5. Determine the reference gain,  $k_r$ , such that the steady state gain is 1!

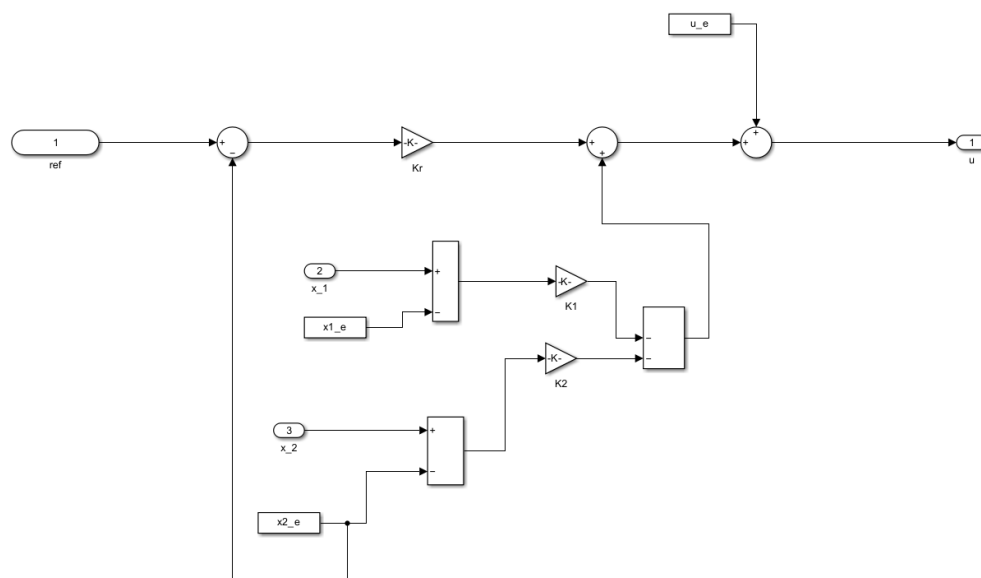
- The solution is provided in the MATLAB. Please refer Matlab.

$k_r = 0.342902822262105.$

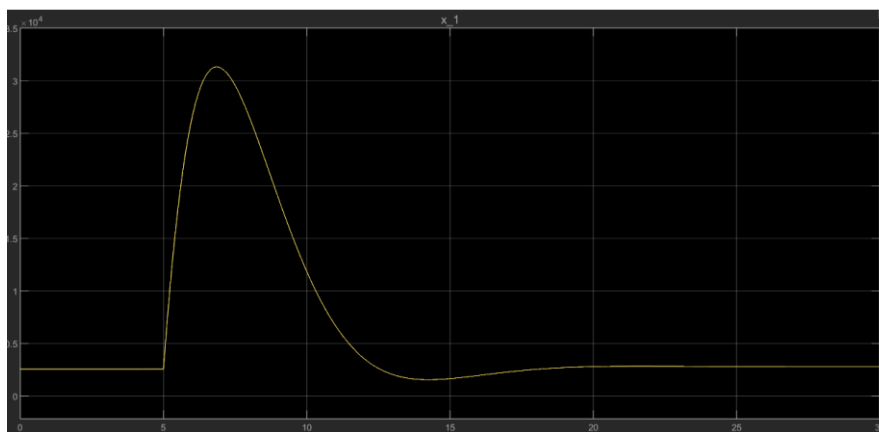
6. Implement the cruise controller! Determine the rise time, the overshoot, and the settling time for the controller when the cruise controllers set speed is changed from 15 m/s to 20 m/s. The rise time  $T_r$  is the time required for the signal to first go from 10% of its final value to 90% of its final value. The overshoot  $M_p$  is the percentage of the final value by which the signal initially rises above the final value. The settling time  $T_s$  is the time required for the signal to stay within 5% of its final value for all future times.

- The code is provided in the MATLAB. Please refer Matlab

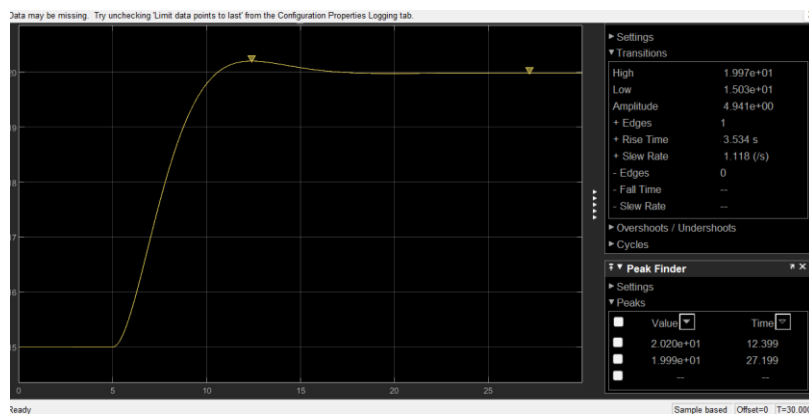
- Refer Simulink model = A3\_model\_cc



Traction force



## Vehicle speed



RiseTime\_6 = 3.577437562979291;

SettlingTime\_6 = 11.691512983610954;

Overshoot\_6 = 4.319346630539322;

7. Augment the system matrices with the additional state variable,  $x_I$ . What will the new system matrices look like? Note that the new state vector will be  $x_a = [x_1 \ x_2 \ x_I]^T$  and the system matrices will have the structure  $\dot{x}_a = A_{aug}x_a + B_{aug}\Delta u + F_{aug}\Delta r$ . Answer with the system matrices,  $A_{aug}$ ,  $B_{aug}$  and  $F_{aug}$ .

Problem 7.  $x_a = \begin{bmatrix} x_1 \\ x_2 \\ x_I \end{bmatrix}$

The integral of the error  $\dot{x}_I = y - \Delta r$ .

$y = x_2 \Rightarrow \dot{x}_I = x_2 - \Delta r$

$$\dot{x}_a = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_I \end{bmatrix} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_I \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} \Delta u + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \Delta r$$

From Problem 2, the A & B matrix where

$$A = \begin{bmatrix} -\frac{1}{T} & 0 \\ \frac{1}{m} & -\frac{P C_D A_f K_{sc}}{m} \end{bmatrix} \quad B = \begin{bmatrix} \frac{K_{rgen}}{T_{rw}} \\ 0 \end{bmatrix}$$

$\therefore$  The Augmented Matrices:  $\dot{x}_a = A_{aug}x_a + B_{aug}\Delta u + F_{aug}\Delta r$

1)  $A_{aug} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{T} & 0 & 0 \\ \frac{1}{m} & -\frac{P C_D A_f K_{sc}}{m} & 0 \\ 0 & 1 & 0 \end{bmatrix}$

2)  $B_{aug} = \begin{bmatrix} B \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{K_{rgen}}{T_{rw}} \\ 0 \\ 0 \end{bmatrix}$

3)  $F_{aug} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

The Augmented system matrices are been found.

8. Design a state feedback controller with integral action using Ackermanns formula for pole placement. Use the same pole placement as before and place the last pole three times faster than the other poles, i.e., place it at  $-3 \zeta \omega_n$ .

- The solution is provided in MATLAB. Please refer Matlab.

$K_{aug} = [0.000079462857141, 2.571858424587528, 0.927433309066289]$

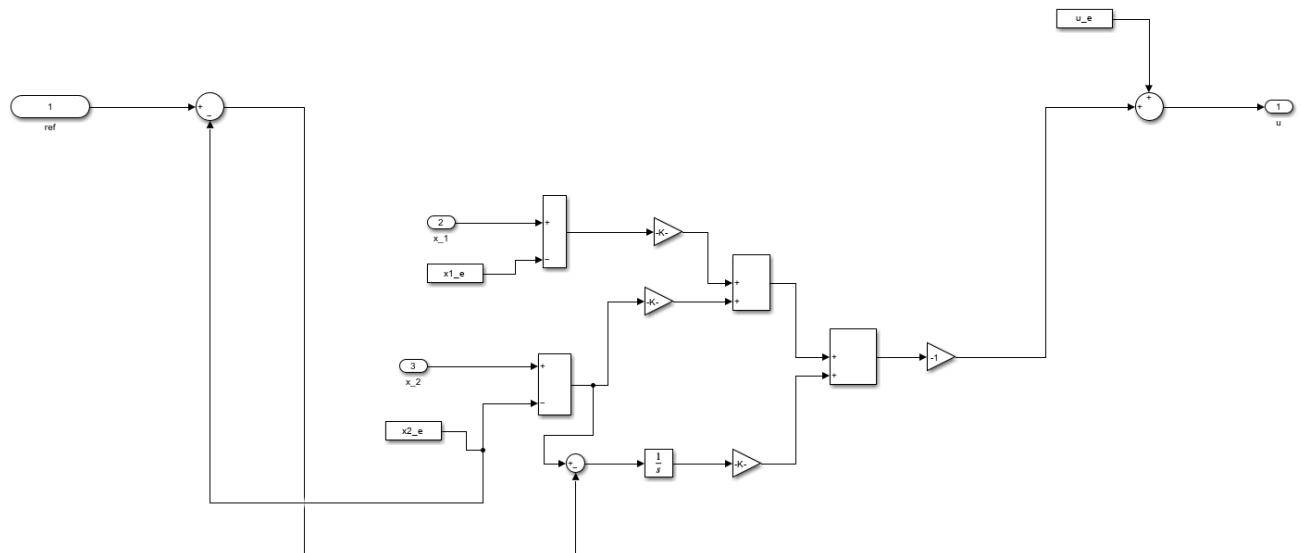
9. Implement the designed cruise controller, including integral action! Determine the rise time, the overshoot and the settling time for the controller when the cruise controller's set speed is changed from 15 m/s to 20 m/s. The rise time  $T_r$  is the time required for the signal to first go from 10% of its final value to 90% of its final value. The overshoot  $M_p$  is the percentage of the final value by which the signal initially rises above the final value. The settling time  $T_s$  is the time required for the signal to stay within 5% of its final value for all future times.

- The code is provided in the MATLAB. Please refer Matlab.

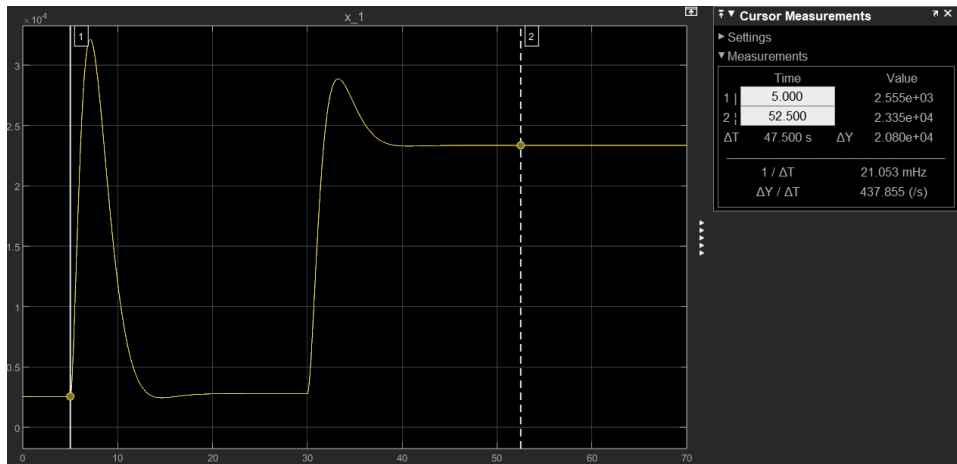
- Refer Simulink model = A3\_model\_cc\_prob\_9

the nominal mass =  $m$  is used in plant model

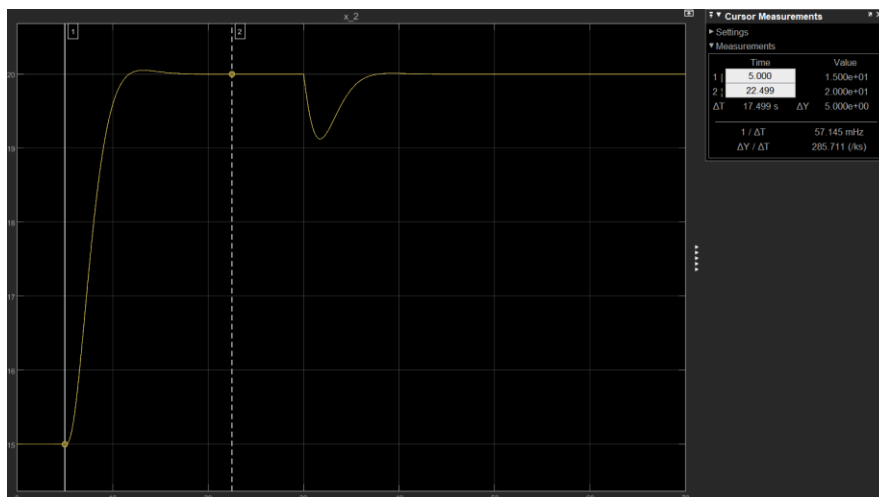
Considering the Stop time = 70sec.



## Traction force



## Vehicle speed



RiseTime\_9 = 3.708820041844949.

SettlingTime\_9 = 29.728305332160414.

Overshoot\_9 = 1.015425253479352.

10. Integral action can sometimes also handle parameter uncertainties. What will happen to the transient performance if the vehicle's mass is increased by 50%? In this problem, we consider a decrease or increase in rise time, overshoot, and settling time compared to the nominal case if the relative change is larger than 5%. Make sure that you only change the mass in the plant model, so your controller design should use the nominal mass.

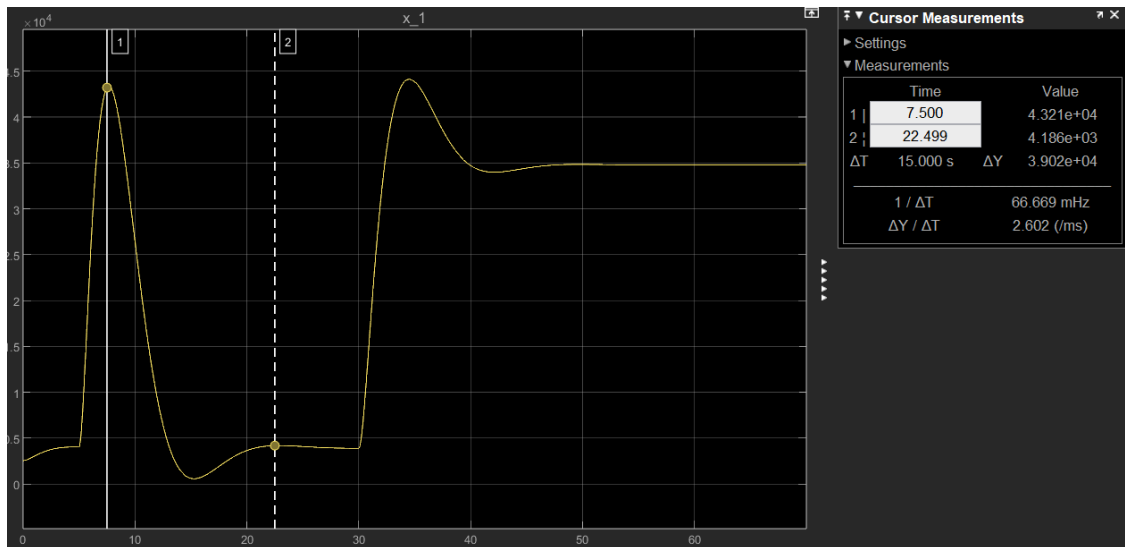
- The solution is provided in the MATLAB. Please refer Matlab.
- Refer Simulink model = A3\_model\_cc\_41

the mass is changed to  $1.5 \cdot m$  in plant model because 50% of mass increase is considered.

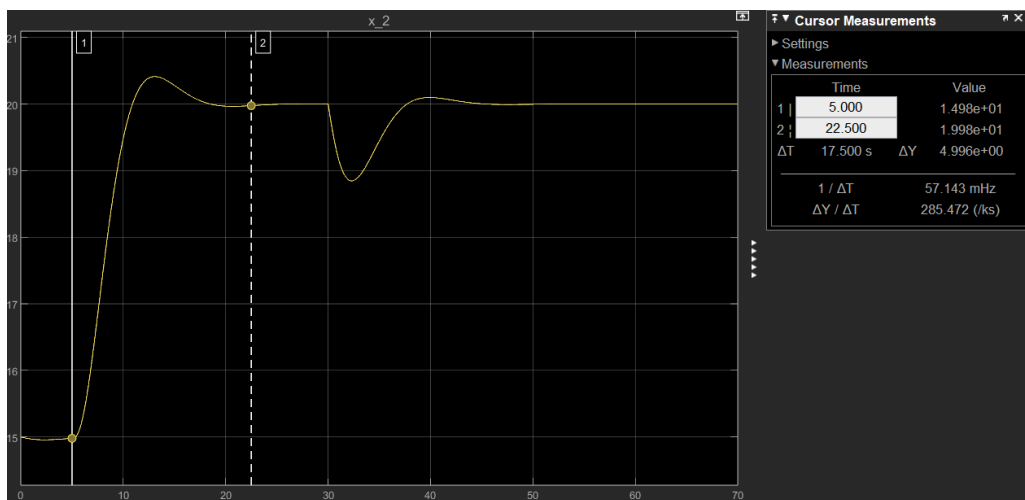
Considering the Stop time = 70sec

Ans\_vec = [0 ; 0 ; 1];                      % [RiseTime, SettlingTime, Overshoot]

Traction force



Vehicle speed



RiseTime\_10 = 3.715625316558095.

SettlingTime\_10 = 31.179099563864760.

Overshoot\_10 = 8.241077564062159.

